



Layer Up Terrain & Trees

Stack flat **fill** layers — one per y

납작한 fill 한 층씩, y를 올리며 쌓아 입체를 만드어요

BIG PICTURE

Flat layers → tall world 🍄



You know the floor (x, z) . New move: **fill** a flat slab, raise **y**, fill again.

바닥(x, z) 위에 fill 한 층씩 y 를 올리면 입체가 돼요

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 **x** & **z** → walk on the ground to pick the **spot** (across + forward).
- 3 **y** → the **layer number**. Add **+1** to stack the next slab on top.

Each **fill** is one flat layer at one **y**. Order is always (x, y, z) .

y는 층 번호 — +1 하면 다음 층이 위에 쌓여요

TYPE IT IN

The **fill** block in Minecraft



^x
across →

^y
layer ↑

^z
forward ↗

```
fill with RED
```

- ✓ Drag the **fill with** block, pick a block colour
- ✓ Set **from** and **to** corners — each is (x, y, z)
- ✓ Same y in both = one flat layer. Raise y for the next one

fill 블록 → from·to 두 모서리 → 둘 다 y가 같으면 한 층, y를 올리면 다음 층

CONCEPT

One fill = one flat layer

Keep **y the same** in **from** and **to** → a flat slab. Bump **y** up by 1 → the next layer sits on top.

```
fill RED from (2, 2, 2) to (10, 2, 10)  # layer y=2  
fill RED from (3, 3, 3) to (9, 3, 9)   # layer y=3
```

from-to의 y가 같으면 납작한 한 층, y를 +1 하면 그 위에 다음 층

CODE

The stem: a small box up

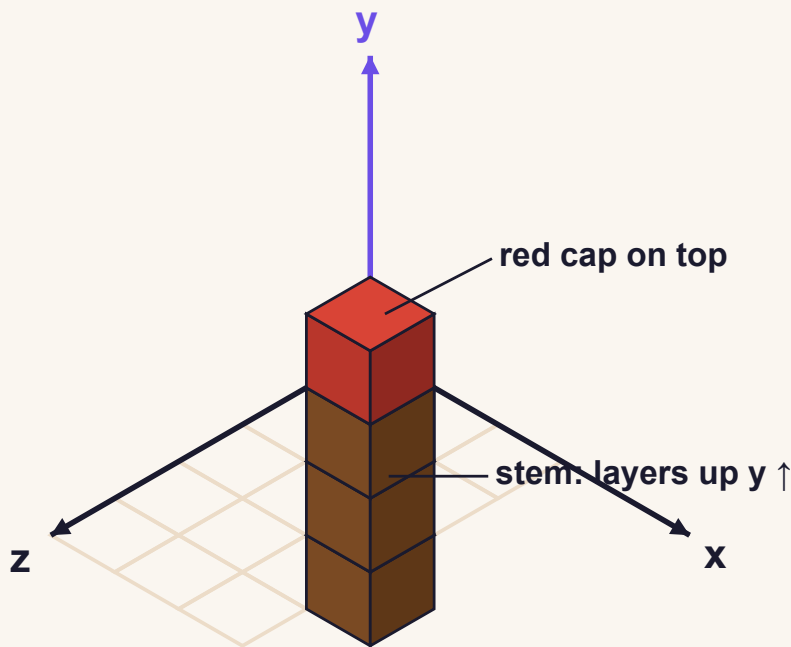
```
fill with BROWN  
  from (4, 0, 4)  
  to   (8, 1, 8) replace
```

From $y = 0$ up to $y = 1$ — a short brown stem 2 layers tall, ready for a cap.

$y = 0 \rightarrow 1$ 까지 채우면 2층짜리 갈색 줄기(기둥)

SEE IT

One spot (x, z) · stacked up (y)



바닥 격자(x, z) 위로 y 가 솟아요 — 갈색 줄기를 쌓고 꼭대기에 빨간 모자를 올리면 버섯!

CODE

A mushroom, layer by layer

```
# stem
fill BROWN from (4,0,4) to (8, 1,8)
# cap – raise y each line, shrink the spread
fill RED    from (2,2,2) to (10,2,10)
fill RED    from (3,3,3) to (9, 3,9)
fill RED    from (4,4,4) to (8, 4,8)
fill RED    from (5,5,5) to (7, 5,7)
```

y climbs 2 → 5. Each layer is a little narrower → a round cap.

y가 2→5로 오르고, 층마다 좁아져서 둥근 모자가 돼요

The Python it makes

```
def on_run():  
    blocks.fill(BROWN_CONCRETE,  
        pos(4, 0, 4), pos(8, 1, 8),  
        FillOperation.REPLACE)          # stem  
    blocks.fill(RED_CONCRETE,  
        pos(2, 2, 2), pos(10, 2, 10),  
        FillOperation.REPLACE)          # cap, y = 2  
    # ...3 more cap layers: y = 3, 4, 5  
    player.on_chat("run", on_run)
```

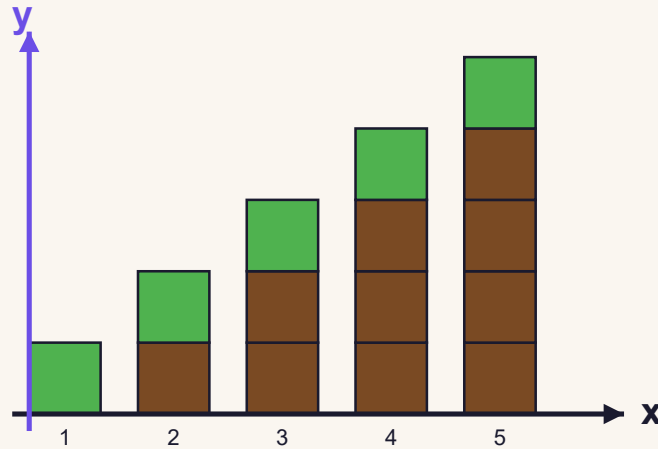
`pos(x, y, z)` is the spot · `REPLACE` overwrites what's there · type **run** in chat to build.

블록을 파이썬으로 보면 — `pos(x, y, z)`가 자리, `REPLACE`는 덮어쓰기. 채팅창에 `run` 입력

CONCEPT

Terrain = layers of different height

side view — each fill stacks to a different y



Move along x. Give each spot a taller top **y**. Bumpy ground = terrain.

x를 따라가며 자리마다 꼭대기 y를 다르게 = 울퉁불퉁한 땅(지형)

CODE

Build a slope of terrain

```
fill DIRT from (0,0,0) to (0,1,5)    # low  
fill DIRT from (1,0,0) to (1,2,5)  
fill DIRT from (2,0,0) to (2,3,5)  
fill DIRT from (3,0,0) to (3,4,5)    # high
```

Each strip stops at a taller **y** — the ground rises into a hill.

줄마다 꼭대기 y가 더 높아져 언덕이 생겨요

PREDICT 🤖

What grows between these layers?

Predict first — then click reveal

```
fill BROWN from (4, 0, 4) to (8, 1, 8)
fill RED    from (2, 2, 2) to (10, 2, 10)
fill RED    from (5, 5, 5) to (7, 5, 7)
```

Reveal output 🔍 reveal

VOCAB

New words today

layer	one flat slab from a single fill
stack	put layers on top by raising y
terrain	ground built up to different heights
cap	the wide top layers of a mushroom/tree

층

쌓다

지형

갓 / 윗부분



What's wrong with this code?

Goal: stack a **second layer on top** of the first.

buggy

```
fill RED from (2, 2, 2) to (10, 2, 10)  
fill RED from (3, 2, 3) to (9, 2, 9)
```

Both layers use $y = 2$. Think first.



DEBUG

The bug

Bug: both fills sit at $y = 2$, so the second lands **on top of** the first — nothing stacks up. Raise the **middle number** to $y = 3$ for the next layer.

fixed

```
fill RED from (2, 2, 2) to (10, 2, 10)
fill RED from (3, 3, 3) to (9, 3, 9)
```

YOUR TURN!

Grow terrain + a mushroom

- ✓ Fill a stem box (a few layers tall)
- ✓ Stack cap layers — raise `y` by 1 each, shrink the spread
- ✓ Make 3 terrain strips that stop at **different heights**
- ✓ Write down the `y` of your top layer
- ✓ Send a photo from the ground up to the top



QUIZ

QUESTION 1 OF 3

A **fill** with the **same y** in from and to makes...

Nothing.

A hole in the ground.

One flat layer at that height.

A tall tower.



QUIZ

QUESTION 2 OF 3

To stack the next layer **on top**, which number do you raise?

y (the middle one)

x (the first one)

z (the last one)

All three the same



QUIZ

QUESTION 3 OF 3

How do you make bumpy **terrain**?

Use the same height everywhere.

Fill one block.

Stop each strip at a different top y.

Only move forward in z.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



You stacked it up!

Next: command all three numbers at once.

다음 주: x , y , z 세 숫자를 한꺼번에